Escaping Saddle Points

Report: October 30, 2026 **Presentation: November 4, 2026**

What you will do

You will study **one benchmark function using three algorithms**: gradient descent (GD), perturbed gradient descent (P-GD), and gradient descent with negative curvature (GD-NC). First compare them for different curvatures and starting points. Then vary their parameters, explain the effects, and propose one algorithmic modification.

The project has three stages: **prepare** by analyzing the function and reading two papers; **experiment** by answering Questions 1–4 and developing your modification; and **explain** your findings and proposed method in a report and presentation. The methods, common setup, and questions appear on the following pages.

Research question. Can your modification improve escape from a saddle? Explain when it helps, when it fails, and what extra computation it requires. A carefully supported negative result can earn full credit.

The benchmark function

For $\gamma > 0$, use

$$f_\gamma(x, y) = \frac{1}{2}x^2 - \frac{\gamma}{2}y^2 + \frac{1}{4}y^4. \tag{1}$$

Implement this formula and its derivatives; no data generation is needed. The parameter γ controls the magnitude of negative curvature at the saddle $(0, 0)$. The experiments section specifies the values of γ and the starting points.

Preparation A1. Understand the geometry

1. Derive the gradient and Hessian, classify all stationary points, and find the minimum value $f_\gamma^\star$. Check your derivatives numerically at nonstationary points.
2. Derive the fixed-step GD updates for $0 < \eta < 2$. Set $a = 0.1\sqrt{\gamma}$ and start with $0 < |y_0| \ll a$. Estimate the iterations needed to reach $|y_k| \geq a$, stating your approximation. Explain what happens if $y_0 = 0$.
3. Derive the undamped Newton update where defined. Explain why it need not escape the saddle. Newton is used only for this analysis.

Preparation A2. Read and compare both papers

Read **both papers in full**, including their theoretical sections:

- C. Jin et al., *How to Escape Saddle Points Efficiently* (ICML 2017).
- J. D. Lee et al., *Gradient Descent Only Converges to Minimizers* (COLT 2016).

For each paper, explain the problem and main contribution, the method or algorithm studied, its assumptions and main guarantee, and the intuition behind one key theoretical argument. Compare the roles of randomness and what the results say about escape time. Finally, explain how our practical P-GD differs from Jin et al.'s Algorithm 2 and why its guarantee does not automatically transfer. Full proof reproduction is not required.

1. The three algorithms

Brief explanations and update rules

Write $z_k = (x_k, y_k)^\top$, $g_k = \nabla f_\gamma(z_k)$, and $H_k = \nabla^2 f_\gamma(z_k)$. Below, f denotes the benchmark function for the chosen γ.

Gradient descent (GD)

GD moves opposite to the gradient. It has no explicit escape mechanism: at an exact stationary point, its update is zero.

$$\boxed{z_{k+1} = z_k - \eta_k g_k}, \qquad f(z_k - \eta_k g_k) \le f(z_k) - 10^{-4}\eta_k \|g_k\|^2 .$$

Choose η_k by backtracking: start at 1 and halve until the inequality holds.

Practical perturbed GD (P-GD)

A small random displacement can introduce motion in an escape direction. Apply it when the gradient norm is below the trigger g_{trig} and at least T updates have passed since the last perturbation. Let ℓ be that iteration, initially $-T$:

$$w_k = \begin{cases} z_k + \xi_k, & \|g_k\| \le g_{\text{trig}} \text{ and } k - \ell \ge T, \\ z_k, & \text{otherwise,} \end{cases}$$

$$\boxed{z_{k+1} = w_k - \eta_k \nabla f(w_k)}.$$

After a perturbation, set $\ell = k$ and recompute the gradient. Apply the GD line search at w_k. To sample uniformly from a disk of radius r, draw independent $v \sim \mathcal{N}(0, I_2)$ and $u \sim \text{Uniform}(0, 1)$, then set $\xi_k = r\sqrt{u}\, v / \|v\|$.

The random displacement itself need not decrease f. A small gradient can also occur near a minimum, so perturbations may disturb a good point.

GD with negative curvature (GD-NC)

Use GD while $\|g_k\| > g_{\text{trig}}$. Otherwise, compute the smallest algebraic eigenpair (λ_k, v_k) of H_k, with $\|v_k\| = 1$. If $\lambda_k < -10^{-4}$, choose $p_k = \pm v_k$ with $g_k^\top p_k \le 0$ and take

$$\boxed{z_{k+1} = z_k + t_k p_k}, \qquad f(z_k + t_k p_k) \le f(z_k) + 10^{-4} t_k^2 \lambda_k.$$

Choose t_k by halving from 1 until the inequality holds; use a fixed sign convention if $g_k^\top v_k = 0$. If the eigenvalue test fails, take the GD step. Negative curvature gives a direction of decrease, but finding it requires Hessian and eigenvalue computations.

Default parameters

Parameter	Meaning	Default
g_{trig}	Small-gradient trigger for P-GD and GD-NC	10^{-3}
r	P-GD perturbation radius	10^{-2}
T	Minimum updates between P-GD perturbations	20

Use $K = 3000$ **updates** per run. Both line searches restart from 1, halve after rejection, and allow 40 trials. Report failure if none is accepted. Keep these line-search rules and GD-NC's curvature threshold fixed throughout.

2. Common setup and results

Use the same rules and one table format throughout

Run and record

Use identical starting points across methods. Run each deterministic method once per setting and each randomized method ten times, with seeds $0, \ldots, 9$. Use the same seed list when changing parameters.

Run all 3000 updates unless a line search fails; do not stop just because the gradient is small. Save $z_0, \ldots, z_K$, cumulative optimizer time, and evaluation counts. A perturbation followed by GD counts as one update; a negative-curvature step replaces that iteration's GD step.

Count function, gradient, and full Hessian evaluations as N_f, N_g, N_H, including rejected line-search trials and gradients recomputed after perturbations. Note any additional curvature work used by your proposed method. Compute the measures below *after the run*; exclude this diagnostic work from optimizer time and counts.

What to measure

Let $f_s = f_\gamma(0,0) = 0$, let $f^\star$ be the minimum found in A1, and set $\Delta = f_s - f^\star > 0$. Use the normalized gap

$$q_k = \frac{f_\gamma(z_k) - f^\star}{\Delta}.$$

- **Escape iteration k_{esc}:** the first k for which $f_\gamma(z_k) \le f_s - 0.1\Delta$. Omit this measure for the near-minimum start.
- **Success iteration k_{succ}:** the first k satisfying all three conditions

$$q_k \le 10^{-4}, \qquad \|\nabla f_\gamma(z_k)\| \le 10^{-5}, \qquad \lambda_{\min}(\nabla^2 f_\gamma(z_k)) \ge -10^{-4}.$$

- **Time to success:** cumulative optimizer time at k_{succ}, in seconds.
- **Final gap and work:** q_K and (N_f, N_g, N_H) for the complete run.

Reaching success and staying there are different. A run succeeds if it meets the tests at least once. Continue the run and report its final gap, since P-GD may perturb away from a minimum later.

One results-table template

Use this format for all experiments. Add one row per method and setting, and additional rows for each parameter value tested. Identify the case and configuration, for example "S2, default" or "S2, $r = 0.05$".

Setting / parameters	Method	Success runs / total	Escape k_{esc}	Success k_{succ}	Time to success (s)	Final gap	Work $N_f/N_g/N_H$
____	GD	____	____	____	____	____	____
____	P-GD	____	____	____	____	____	____
____	GD-NC	____	____	____	____	____	____
____	Proposed	____	____	____	____	____	____

For repeated runs, report **mean ± sample standard deviation** in the numerical columns. Summarize escape over runs that escape and success iteration/time over successful runs. If not all runs escape, include the escape count. Use "NR" for unmet targets and "–" when escape is not applicable. If only one value is available, report it without a standard deviation. Keep success as a count out of 1 or 10 runs.

Report final gaps and work over all runs. If a line search fails, use the last accepted iterate for the final gap and note the failure and completed updates. Keep individual-run values with your code.

3. Experiments: Questions 1–4

Compare the methods and check parameter effects

Use the common setup on page 3 and fill the results table as you work. Questions 1–3 use GD, P-GD, and GD-NC with their **default parameters** in the following settings.

Case	γ	Starting point (x_0, y_0)	What changes?
S1	1	$(0.05, 10^{-8})$	Near the saddle
S2	0.1	$(0.05, 10^{-8})$	Weaker negative curvature
S3	0.01	$(0.05, 10^{-8})$	Still weaker negative curvature
S4	1	$(0, 0)$	Exactly at the saddle
S5	1	$(0.05, 1 + 10^{-4})$	Near a minimum

Question 1. How does curvature affect escape?

Purpose: understand the effect of weaker negative curvature.

Run **S1**–**S3**: $\gamma = 1, 0.1, 0.01$ with the same near-saddle start. Compare escape, success, accuracy, and computation. Explain the trends using your analysis in A1.

Question 2. What happens at the exact saddle?

Purpose: distinguish escaping an exact saddle from moving away from a nearby point.

Run **S4**: $\gamma = 1$ and $(x_0, y_0) = (0, 0)$. Compare with S1 and explain which update creates motion away from the saddle. Report any failure to escape.

Question 3. What happens near a minimum?

Purpose: check whether an escape mechanism disturbs a good point.

Run **S5**: $\gamma = 1$ and $(x_0, y_0) = (0.05, 1 + 10^{-4})$. Omit escape time. Compare first success, final gap, and computation. Explain any unnecessary perturbations.

Question 4. What is the effect of the parameters?

Purpose: understand which choices help escape and which increase cost or reduce final accuracy.

Use **S2**. Vary **one parameter at a time**, keeping the others at their defaults. Test at least three distinct values for each row, including the default and both ends of the range. The three suggested values meet this requirement.

Method	Parameter	Range	Suggested values
P-GD, GD-NC	g_{trig}	$[10^{-4}, 10^{-2}]$	$10^{-4},\ 10^{-3},\ 10^{-2}$
P-GD	r	$[10^{-3}, 5 \times 10^{-2}]$	$10^{-3},\ 10^{-2},\ 5 \times 10^{-2}$
P-GD	T	Integers in $[10, 50]$	$10,\ 20,\ 50$

Describe the effect on escape, success, final gap, and computation. Choose one configuration for P-GD and one for GD-NC from those you actually tested, and briefly justify each choice. Reuse the default runs; GD has no parameter to vary here.

Figures: include S2 contours with trajectories and perturbations marked, S2 normalized gaps versus updates, and one plot showing a parameter effect from Question 4. Include your proposed method in the S2 comparison plots; label the seed of any illustrated randomized run. Use the table to summarize all runs.

4. Proposed method and submission

Explain your contribution and findings

Develop and evaluate your proposed method

Address a limitation observed in Questions 1–4 and explain the effect of your change.

1. **Propose and explain.** Change an update or decision rule; changing fixed parameter values alone is insufficient. Give equations or pseudocode, state your parameters, and justify the idea with a short mathematical argument. Distinguish proved claims from conjectures; publication-level novelty is not required.
2. **Compare.** Test your method in S1–S5 against GD and the two configurations chosen in Question 4. State your method's configuration before this comparison and keep every method's parameters fixed across the five cases. Reuse any unchanged baseline runs.
3. **Check the effect of your modification.** In S2, compare your method with the original algorithm using the same remaining parameters, starting point, and random seeds. Explain the benefits, additional computation, and limitations. Use your earlier baseline results when the settings match.

Deadlines and deliverables

Deadline	Submission
October 30, 2026	**Final report and accompanying code.**
November 4, 2026	**Presentation and discussion.**

Report

Submit a **professionally written report prepared in LaTeX**, with clear figures, tables, and detailed explanations of your analysis, methods, and results.

Presentation

Discuss your findings and proposed method. Be prepared to answer personalized questions about the papers, your mathematical reasoning, implementation, and interpretation of the results.

Assessment: 100 points

Component	Points
Understanding the two papers	20
Questions 1–4	20
Report	10
Proposed method	25
Presentation	25